SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 1 – Research Paper Review / Literature Survey

Course Code:	**CSA1008**	**Course Title:**	**Software Engineering**
CO Assessed:	**CO5**	**Assessment:**	**Assessment Tool 1 – Research Paper Review / Literature Survey**
Weightage:	**20%**	**Total Marks:**	**25**
CO5 Statement:	**Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly**		
Student Name:	**B.Pushparaj**	**Reg. No.:**	**192421136**
Date:	**26.08.2026**	**Duration:**	**60 minutes**

Code of Conduct

I **B. Pushparaj (192421136)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _ 

____

Annexure A – Research Paper Review / Literature Survey

Instructions to Students:

Each student is assigned an individual problem statement. Based on the assigned problem statement, conduct a literature survey and review relevant research papers to understand the existing approaches, techniques, technologies, limitations, and research gaps related to the problem.

The literature survey should include:

1. Identify and select relevant research papers related to the assigned problem statement from reputed journals, conferences, or scholarly databases.
2. Summarize the objectives, methodologies, technologies, datasets, and major findings of the selected research papers.
3. Compare the existing approaches based on suitable parameters such as accuracy, performance, scalability, efficiency, methodology, or other problem-specific metrics.
4. Identify the limitations and challenges of the existing research approaches.
5. Analyze the literature to identify the research gap or unresolved issues related to the assigned problem.
6. Discuss the possible improvements or future research directions based on the identified research gap.
7. Prepare a comparative literature survey table containing details such as author/year, research problem, methodology, dataset/tools, key findings, limitations, and research gap.
8. Provide proper citations and references using a consistent referencing style.

Deliverable: Submit a Research Paper Review / Literature Survey Report containing the selected research papers, individual paper reviews, comparative analysis, identified research gaps, future directions, and properly formatted references.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Selection & Review of Research Papers(25%)	Selects highly relevant, recent, credible papers and provides comprehensive reviews of objectives, methodology, tools, datasets, and findings	Selects relevant papers and provides good reviews with minor gaps	Selects moderately relevant papers with basic reviews	Papers are irrelevant or reviews are incomplete
Comparative Analysis(20%)	Provides a detailed comparison of methodologies,	Provides a good comparison with relevant parameters	Provides limited comparison	Little or no meaningful comparison

	tools, datasets, results, and performance using a clear comparison table		with few parameters	
Critical Analysis & Research Gap(25%)	Clearly analyzes limitations and identifies significant, well-supported research gaps	Identifies relevant limitations and research gaps	Identifies basic gaps with limited analysis	Research gaps are unclear or missing
Future Directions & Relevance(15%)	Proposes practical and innovative future directions directly relevant to the assigned problem statement	Proposes relevant future directions	Provides general future suggestions	Future directions are missing or unrelated
Documentation, Citations & References(15%)	Excellent academic structure, clear presentation, accurate citations, and consistent referencing	Well-organized with minor formatting/citation issues	Adequately documented with some errors	Poorly organized with inadequate or incorrect references

Criteria	**Mark**
Selection & Review of Research Papers(25%)	**/5**
Comparative Analysis(20%)	**/5**
Critical Analysis & Research Gap(25%)	**/5**
Future Directions & Relevance(15%)	**/5**

Documentation, Citations & References(15%)	**/5**
TOTAL	**/25**

CO5 – Research Paper Review / Literature Survey

Project Topic

Accessible Learning Platform for Students with Visual Impairments

1. Introduction

The **Accessible Learning Platform** is designed to provide an inclusive digital learning environment for students with visual impairments. Conventional online learning platforms may contain accessibility barriers such as images without alternative text, difficult navigation, inaccessible assessments, and poor compatibility with screen readers.

The proposed platform focuses on **screen-reader support, voice navigation, accessible learning materials, audio resources, keyboard navigation, and accessible assessments**.

Research shows that accessibility problems continue to exist in online learning platforms. Studies have reported issues including missing alternative text, poor keyboard navigation, unclear links, missing form labels, and insufficient support for assistive technologies. (Springer)

2. Objectives of the Literature Survey

The main objectives are:

1. To study existing accessible e-learning platforms.
2. To identify accessibility techniques used for visually impaired students.
3. To compare existing research approaches.
4. To identify limitations in existing systems.
5. To find the research gap.
6. To propose improvements for the Accessible Learning Platform.

3. Research Papers Selected

For the literature survey, the following research works are relevant to the project:

No.	Research Area	Main Focus
1	Accessible MOOCs	Accessibility of online learning platforms
2	Accessible Online Learning	Inclusive learning in higher education
3	Accessible Mathematics Platform	E-learning for blind and visually impaired students
4	OER Accessibility	Accessibility of Open Educational Resources
5	Digital Accessibility	Problems faced by visually impaired users
6	Mobile E-learning	Mobile learning for visually impaired children
7	Higher Education Websites	Accessibility problems in university websites

4. Individual Research Paper Review

Paper 1 – Accessibility of MOOCs

Research Focus:
The study investigated the accessibility of MOOCs for users with disabilities.

Methodology:
The researchers evaluated online courses and collected feedback from users with visual and movement impairments.

Technologies/Tools:

- Screen readers
- Keyboard navigation
- WCAG 2.1 guidelines
- Web-based learning platforms

Major Findings:
Problems were found with alternative text, keyboard operation, video controls, links, labels, and focus visibility. (Springer)

Limitation:
The study mainly evaluates existing platforms rather than creating a complete accessibility-focused learning system.

Relevance to Our Project:
Our system can directly address these problems through accessible navigation, proper labels, alternative content and screen-reader support.

Paper 2 – Accessible and Inclusive Online Learning

This research reviewed **91 sources** related to accessible and inclusive online learning in higher education. It identified accessibility of course content, multimedia, navigation and assistive technology compatibility as important issues. (Springer)

Key Finding:
Online learning can support students with disabilities, but inaccessible content can become a major barrier.

Limitation:
The work is primarily a literature review and does not provide one complete implementation framework.

Relevance:
Our project combines accessibility features into one learning platform.

Paper 3 – Multimedia Learning Platform for Blind Students

This study developed an interactive mathematics learning platform accessible to blind and visually impaired students.

The platform was evaluated with **23 students with impairments**, including blind, low-vision and motor-impaired users. Screen readers and assistive technologies were considered during evaluation. (Springer)

Key Finding:
Accessible learning platforms can provide effective learning experiences when designed around assistive technologies.

Limitation:
The platform is focused mainly on mathematics education.

Research Opportunity:
Our project can provide a **general-purpose learning platform** rather than focusing on one subject.

Paper 4 – Accessibility of Open Educational Resources

A systematic literature review studied accessibility within Open Educational Resources and Open Educational Practices. The review found that many educational resources still require improvement in accessibility. (Springer)

Important Technologies/Standards:

- WCAG
- Accessibility metadata
- Alternative content
- Assistive technologies

Limitation:
Many existing approaches do not provide complete compatibility across different assistive technologies.

Relevance:
Our system can provide accessible resources using a consistent accessibility design.

Paper 5 – Digital Accessibility for Visual Impairment

A scoping review analyzed research concerning digital technologies used by people with visual impairment and blindness. It identified problems related to visual-only information, screen-reader compatibility, text size, colors and interface design. (Springer)

Key Finding:
Digital systems should not depend only on visual information.

Limitation:
The study identifies accessibility problems across many types of digital systems rather than implementing a specific educational platform.

Relevance:
Our project focuses these accessibility principles specifically on education.

Paper 6 – Mobile E-Learning for Visually Impaired Students

A mobile e-learning application was developed to improve mathematical skills among visually impaired children. The work demonstrates how mobile technology can support accessible education. (Springer)

Limitation:
The application is focused on basic mathematics and mobile learning.

Improvement in Our Project:
Our platform can support multiple subjects, learning materials, assessments and teacher/student interaction.

5. Comparative Literature Survey

Research	Methodology	Technology	Main Finding	Limitation
Accessible MOOCs	User evaluation	Screen reader, WCAG	Several accessibility barriers identified	Mainly evaluation
Inclusive Online Learning	Literature review	WCAG, UDL	Accessible content is important	No complete implementation
Mathematics Platform	User testing	Screen readers, web	Supports blind students	Subject-specific
OER Accessibility	Systematic review	WCAG, metadata	OER accessibility needs improvement	Limited assistive-tech coverage
Digital Accessibility	Scoping review	Assistive technologies	Visual-only information creates barriers	Broad digital scope
Mobile E-learning	Application development	Mobile technology	Supports visually impaired learners	Mainly mathematics
Higher Education Websites	Accessibility analysis	WCAG	Universities have accessibility problems	Website-focused

6. Existing System Limitations

From the literature survey, the following common problems are identified:

1. Lack of proper screen-reader support.
2. Images may not contain alternative text.
3. Difficult keyboard navigation.
4. Inaccessible online assessments.
5. Video/audio content may not have suitable alternatives.
6. Poor form labels and navigation structure.
7. Dependence on visual information.
8. Limited personalization for different accessibility needs.
9. Accessibility is often treated as an additional feature rather than a core design requirement.

These issues have been reported across research on MOOCs, OER and higher-education websites. (Springer)

7. Research Gap

The literature indicates that many studies focus on **evaluating accessibility** or developing accessibility solutions for a specific type of learning.

However, there is an opportunity to develop a **single integrated learning platform** that combines:

- Screen-reader-friendly interface
- Voice navigation
- Audio learning resources
- Accessible documents
- Keyboard navigation
- Accessible quizzes and examinations
- Alternative text for images
- Teacher content management
- Student progress tracking
- Accessibility-focused UI design

Therefore, the major research gap is:

Existing studies provide individual accessibility techniques or evaluate existing platforms, but there is a need for an integrated learning platform that combines multiple accessibility features into one simple educational environment.

This gap is consistent with reviews reporting continuing barriers across online learning content and assistive-technology compatibility. (Springer)

8. Proposed Solution

Our proposed **Accessible Learning Platform** provides a centralized environment for students and teachers.

Student Module

- Student login
- View courses
- Access learning materials
- Listen to audio content
- Voice navigation
- Take accessible quizzes
- View marks and progress

Teacher Module

- Teacher login
- Upload learning materials
- Add audio/video resources
- Create accessible quizzes
- View student performance

Accessibility Module

- Screen-reader compatibility
- Keyboard navigation
- Voice commands
- Alternative text
- Adjustable text/interface
- Accessible assessment interface

9. Proposed Architecture

```
      STUDENTS / TEACHERS
              |
              ▼
      Web / Mobile Interface
              |
              ▼
       Application Server
              |
      ┌───────┴───────┐
      |       |
Voice Services   Learning Module
      |       |
      └───────┬───────┘
              ▼
        MySQL Database
```

Architecture Explanation

Web/Mobile Interface:
Provides the main interface for students and teachers.

Application Server:
Handles login, courses, assessments and user requests.

Voice Services:
Provides voice navigation and audio-based interaction.

Learning Module:
Manages courses, materials, quizzes and student progress.

MySQL Database:
Stores user, course, assessment and progress information.

10. Proposed Innovation

The main contribution of our project is not simply creating another online learning website.

Our focus is:

Accessibility-first learning design.

Instead of adding accessibility after development, accessibility features are considered from the beginning of system design.

Key Innovation

“One platform + multiple accessibility methods.”

A student can access learning content through:

Text → Screen Reader → Keyboard → Voice → Audio

This gives students more than one way to interact with the same educational content.

11. Future Research Directions

Future improvements can include:

1. Support for additional disabilities.
2. Multilingual voice navigation.
3. Offline accessible learning.
4. Braille-display compatibility.
5. Personalized accessibility settings.
6. Mobile application development.
7. Automatic accessibility checking of uploaded content.
8. Integration with existing college LMS systems.
9. Improved accessible examination systems.
10. Continuous accessibility testing using WCAG-based tools.

Recent research also emphasizes the need for continued accessibility auditing and improvement of digital educational resources. (Springer)

12. Conclusion

The literature survey shows that **accessibility remains an important challenge in digital education**, particularly for visually impaired students. Existing research has demonstrated the importance of screen readers, keyboard navigation, alternative text, accessible multimedia and WCAG-based design.

However, many existing solutions are either evaluation-based or limited to particular subjects, platforms or accessibility features.

Therefore, the proposed **Accessible Learning Platform** aims to integrate multiple accessibility mechanisms into one learning environment. The system can help students access educational content more independently while also helping teachers create and manage accessible learning resources.

13. References

1. Accessible and inclusive online learning in higher education: A review of the literature. *Journal of Computing in Higher Education*. (Springer)
2. Analysis of the accessibility of selected massive open online courses (MOOCs) for users with disabilities. *Universal Access in the Information Society*. (Springer)
3. Multimedia platform for mathematics' interactive learning accessible to blind people. *Multimedia Tools and Applications*. (Springer)
4. Accessibility within open educational resources and practices for disabled learners: A systematic literature review. *Smart Learning Environments*. (Springer)
5. The accessibility of digital technologies for people with visual impairment and blindness: A scoping review. *Discover Computing*. (Springer)
6. A mobile e-learning application for enhancement of basic mathematical skills in visually impaired children. *Universal Access in the Information Society*. (Springer)
7. Web accessibility investigation and identification of major issues of higher education websites. *Journal of King Saud University Computer and Information Sciences*. (Springer)
8. The use of accessibility metadata in e-learning environments: A systematic literature review. *Universal Access in the Information Society*. (Springer)
9. Media Accessibility in E-Learning: Analyzing Learning Management Systems. *Springer*. (Springer)
10. Effects of interactive and hands-on learning experience of assistive technology on college students' perspectives and knowledge. *Education and Information Technologies*, 2026. (Springer)

This structure directly covers the CO5 rubric: paper selection, individual review, comparison, limitations, research gap, future directions, documentation, citations and references.